Question:
At birth, the least developed part of the brain is the
Answer Choices:
A. visual system
B. thalamus
C. frontal lobe
D. brain stem
E. parietal lobe
F. cerebellum
G. cortex
H. basal ganglia
I. hypothalamus
J. limbic system

Reasoning + Answer:

</tool_call> reasoning process here
At birth , the brain undergo es rapid development , with certain regions developing earlier than others . The brain stem , which controls basic life -s ust aining functions like breathing and heart rate , is one of the first structures to develop and is largely present at birth . The visual system , th alam us , hypoth alam us , limb ic system , basal gang lia , and cere bell um develop later , with the cerebral cortex and especially the frontal l obe being the last to mature .

Among the options , the least developed part of the brain at birth is the cerebral cortex , particularly the frontal l obe , which is responsible for higher cognitive functions and is the most developed later in life . The cortex is not fully my el inated or organized at birth and undergo es significant development post nat ally .

Thus , the least developed part at birth is the cortex , specifically the frontal l obe , which is not mature in structure or function .

A . visual system âĢĵ relatively developed at birth , though mat uration continues
B . th alam us âĢĵ developed early
C . frontal l obe âĢĵ least developed at birth , develops over months and years
D . brain stem âĢĵ developed early
E . par iet al l obe âĢĵ develops post nat ally
F . cere bell um âĢĵ develops steadily over time , mature later
G . cortex âĢĵ includes frontal l obe ; least developed at birth
H . basal gang lia âĢĵ develops gradually
I . hypoth alam us âĢĵ present at birth but functional mat uration is delayed
J . limb ic system âĢĵ develops progressively

Among the options , the frontal l obe (part of the cortex) is the most prominent in terms of maturity delay , making it the least developed at birth .

So the best choice from the list that reflects the most under develop ed region at birth is the frontal l obe (which is part of the cerebral cortex).

However , " c ortex " as a whole is widely recognized as developing last . But the question asks for the " least developed part ", and the frontal l obe is the last region to mature , critically for executive functions .

So the answer is either G (c ortex) or C (front al l obe). However , the " c ortex " as a structure is broad , and within it , the frontal cortex is especially under develop ed at birth . Even so , note that the cortex is the outer layer , and although it develops last , the cortex as a whole is less formed at birth .

But , critical evidence in pediatric neuro development shows the frontal l obe is the last brain region to develop and its mat uration takes until early adolescence .

Among the options , both G and C are present , and the frontal l obe is a region of cortex .

But the option C âĢĶ " front al l obe " âĢĶ is specifically known to be the most immature at birth , especially for pre front al functions .

Final decision : There is some overlap , but the newest neuro development al research shows the frontal l obe , not the whole cortex , is the most under develop ed . Also , the cortex is structural term ; the frontal l obe is a region ; and the frontal l obe is the actual structure mat uring last .

Therefore , the " least developed part " is the frontal l obe .

< answer > C </ answer >